

Measurement of interparticle force between nematic colloids

Yasuyuki Kimura* and Kuniyoshi Izaki

Department of Physics, Kyushu University, 6-10-1 Hakozaki, Higashi-ku, Fukuoka, Japan 812-8581

ABSTRACT

Micro-sized colloidal particles dispersed in nematic liquid crystal become topological defects in uniform orientation of liquid crystal. Since they increase the elastic energy of the liquid crystal, a long-ranged anisotropic interaction is induced between them. In this study, we reported the interparticle force measured by various methods utilizing optical tweezers. The interparticle force depends on the type of particle-defect pair and its dependence on the interparticle distance is in agreement with the theoretical prediction using electrostatic analogy. This anisotropic force enables us to construct characteristic clusters, which cannot be realized in conventional water-based colloidal dispersions. We made some novel colloidal assemblies in two dimensions by utilizing optical tweezers to demonstrate the availability of the anisotropic force in nematic colloids

Keywords: colloids, liquid crystal, interparticle force, optical tweezers, colloidal assembly

1. INTRODUCTION

The interaction between spherical colloids in isotropic liquids has intensively been studied from the fundamental interests and the practical interests to build micro-structures such as photonic crystals. For recent decades, the various kinds of specific interaction between colloids in complex structured fluids has also been studied. The colloidal dispersion in nematic liquid crystal is one of the typical examples of such systems and is called “nematic colloid”¹⁻⁵. Due to the orientational order in nematic liquid crystal, the induced interaction between them is anisotropic and long-ranged compared with the conventional one such as van der Waals and Coulombic ones.

There are different types of nematic colloids depending on the alignment of liquid crystal at particles’ surface, the particle’s size and the anchoring strength at the surface³⁻⁵. Three types of nematic colloids discussed in this study are schematically drawn in Fig. 1. In a dipole colloid (D-particle), a point defect appears nearby the particle (Fig. 1 (a)). In a Saturn-ring colloid (S-particle), a ring defect appears around the particle (Fig. 1 (b)). The normal of the ring is parallel to the alignment of liquid crystal without particles. In a planar quadrupole colloid (P-particle), two surface defects called “boojums” appear at the poles of the particle (Fig. 1 (c)). The line connecting the boojums is parallel to the alignment of liquid crystal. According to the electrostatic analogy of nematic field^{2,3,6}, D-particle interact with other particles via

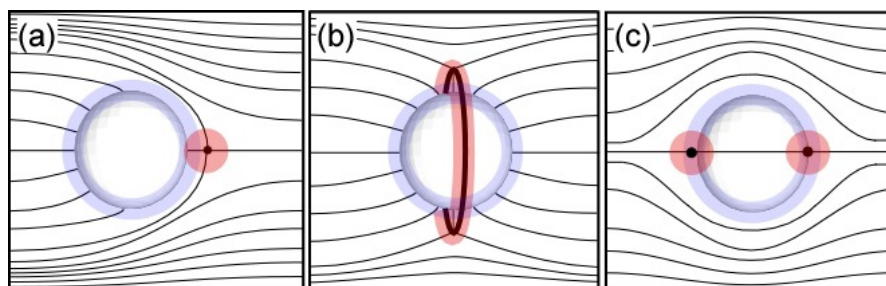


Figure 1. Typical nematic colloids. (a) dipole-colloid (D): the direction of liquid crystal at the particle’s surface is normal to the surface. A point defect appears nearby the particle. (b) Saturn-ring colloid (S): the ring defect is above equator of the particle. (c) planar-quadrupole colloid (P): two surface defects appear at the poles of the particle. The lines in the figures represent the local alignment of liquid crystal.

*kimura@phys.kyushu-u.ac.jp; phone 81 92 642 2534; fax 81 92 642 2534

dipolar interaction. The S- and P-particles interact with other ones via quadrupolar interaction but their signs are opposite.

There are various kinds of measurement methods for interparticle force between nematic colloids. In this study, some new measurement methods those are slightly modified version of the conventional ones are demonstrated and compared with others. The interparticle force between the nematic colloids depicted in Fig. 1 has been studied. We also discuss the availability of this anisotropic interparticle force for building exotic assemblies those have not be realized in isotropic colloids including dendric clusters and empty crystals by manipulating the colloids with optical tweezers.

2. MEASUREMENT METHODS OF THE INTERPARTICLE FORCE BETWEEN NEMATIC COLLOIDS

Frequently used measurement method of the interparticle force between nematic colloids is so-called free-released (FR) method. In FR method, the force is evaluated from the approaching velocity of two particles (Fig.2 (a)). At the stationary state, the interparticle force F balances the drag force in liquid crystal F_{drag} . If one measures the effective viscosity of liquid crystalline medium beforehand, F can be evaluated from the temporal change of the interparticle distance r ^{7,8}. The other method is one by utilizing the magnetic⁹ or optical tweezers¹⁰⁻¹³. In the optical tweezing method^{12,13}, two particles are respectively trapped by dual beam optical tweezers (OT) and the displacement from the center of the trap is measured as a function of r (Fig. 2 (b)). If one measures the stiffness of the optical trapping beforehand, F can be evaluated from the optical trapping force F_{opt} .

In this study, we dispersed polystyrene spherical colloidal particles with radius $a=2.55\pm0.1\mu\text{m}$ (Magsphere Inc.) in a nematic liquid crystal MJ032358 (Merck Japan). Since the refractive index of the particle is larger than that of the liquid crystal, the particles can stably be trapped by optical tweezers. We used a Nd:YVO₄ laser (Spectra Physics, wavelength: 1064 nm) for optical tweezers. The beam was introduced into an inverted microscope (TE2000U, Nikon) and focused using a 100x oil immersion objective lens (Plan Flour, N.A.=1.3, Nikon). The effective viscosity of the nematic liquid crystal parallel and perpendicular to the alignment of the liquid crystal has been evaluated from the diffusion constant obtained by the temporal change of the mean square displacement of the particles by using Stokes-Einstein relation^{12,14}. The stiffness of the optical trapping is estimated from the distribution of the trapped particle's position by assuming Boltzmann distribution¹².

The interparticle force F obtained by FR and OT methods are shown in Fig. 3. The dependence of F on r qualitatively similar to each other: the magnitude of F at large r increases with decrease of r , and decreases at small r . F shows a maximum value at similar r . However, these two force curves exhibit apparent large difference in magnitude. In FR method, the movement of the particles induces flow around them and this reduces the velocity of the counter approaching particle. On the other hand, in OT method, two particle are fixed and the flow induced by the fluctuation of the particles is much smaller. The influence of the hydrodynamic interaction between the particles is also observed in the force curves obtained by other methods, which we called FR-OT method. In FR-OT method, one particle is trapped by optical tweezers and the other is freely released (Fig. 2 (c)). Since the counter particle is fixed, the hydrodynamic interaction becomes smaller for the free-release particle. We can confirm this expectation in Fig. 3.

The interparticle forces obtained by four different methods are largely different from each other¹⁵. In order to correct the measured force, we take into account the hydrodynamic interaction by utilizing the Oseen approximation which has been

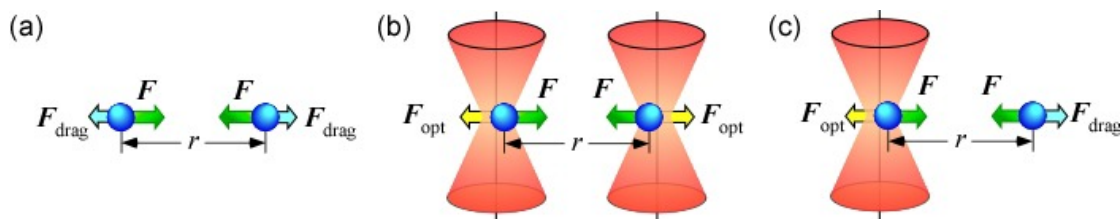


Figure 2. Measurement methods of interparticle force between nematic colloids utilized in this study. (a) free-release method (FR): two particles move freely. (b) optical trapping method (OT): two particles are respectively trapped optical tweezers. (c) free-release-trapping method (FR-OT): one particle is trapped by optical tweezers and the other is free-released. F is the interparticle force (we assume attractive force in the figures), F_{drag} is the drag force, F_{opt} is the trapping force by optical tweezers and r is the interparticle distance.

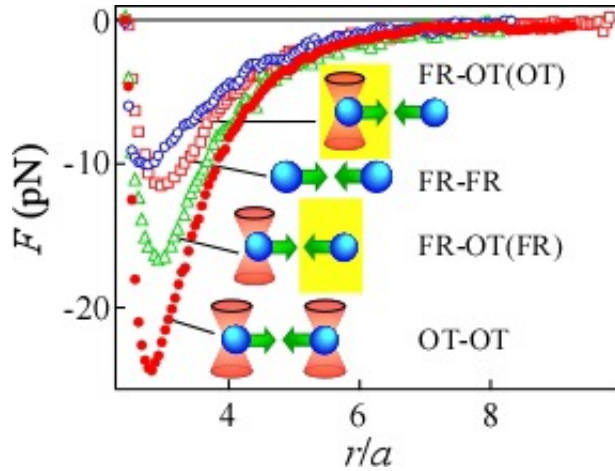


Figure 3. Dependence of the interparticle force F on the reduced interparticle distance r/a measured by various methods. In FR-OT method, the force curves are respectively evaluated from the particle with shade.

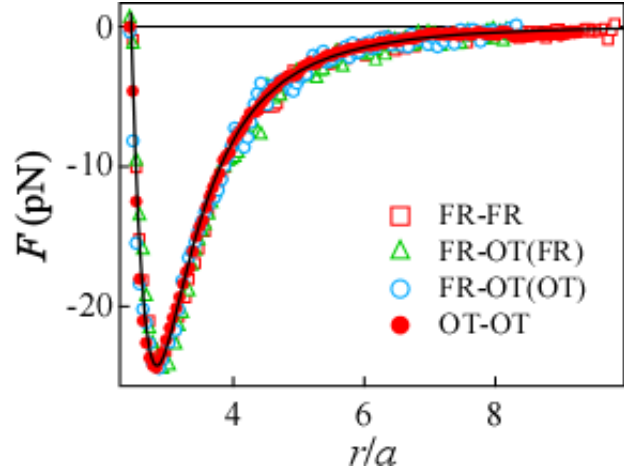


Figure 4. Dependence of the corrected interparticle force F on the reduced interparticle distance r/a measured by various methods. The solid line is the theoretical one by adjusting the maximum magnitude to the experimental data.

well studied in simple fluids with low Reynolds number¹⁶. The hydrodynamic force $F_{h(i,j)}$ that the j -th particle receives from the i -th particle is given by

$$F_{h(i,j)} = \frac{3a}{4r_{ji}} \left[\mathbf{I} + \frac{\mathbf{r}_{ji} \otimes \mathbf{r}_{ji}}{r_{ji}^2} \right] \mathbf{F}_i, \quad (i, j=1,2) \quad (1)$$

where a is the radius of the particles, $\mathbf{r}_{ji} = \mathbf{r}_j - \mathbf{r}_i$, $|\mathbf{r}_{ji}| = r_{ji} = r$. The corrected three force curves in Fig. 3 are shown in Fig. 4. We find the quantitative agreement between all force curves obtained by different methods. The applicability of our simple assumption in evaluation process of F and the validity of neglecting short-range hydrodynamic repulsion has to be checked by more rigorous calculation in future.

3. INTERPARTICLE FORCE BETWEEN VARIOUS NEMATIC COLLOIDS

According to the electrostatic analogy, the interparticle force between D-particles is dipole-dipole one^{2,6} and F is proportional to $(r/a)^{-4}$. The dependence of F on r obtained by experiment is $r^{-4.1}$ at large r . The solid line in Fig. 4 is the theoretically calculated one using elastic theory¹⁷ for two particles align in-line drawn by adjusting the maximum force in magnitude to experimental one. The agreement is rather good in wide range of r/a . At small r , the repulsive component appears and the magnitude of attractive force decreases. The main origin of this repulsive component is the

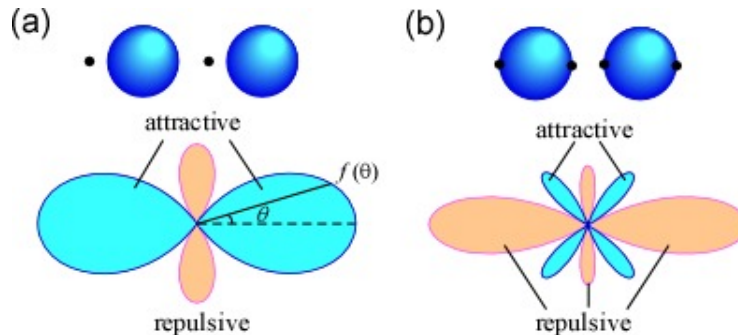


Figure 5. Relative angular dependence of interparticle force $f(\theta)$ (in arbitrary units) between nematic colloids, where θ is the angle between the line connecting the center of two particles and the alignment of liquid crystal. (a) dipole-dipole. (b) quadrupole-quadrupole.

steep change in the orientation of the liquid crystal between the particles. The sign and magnitude of F also depends on the relative arrangement between two dipoles as shown in Fig. 5 (a). By using OT method, we can measure the dependence of F on the relative angle between the particles. We have confirmed the variation of the magnitude and sign of F as predicted in Fig. 5 (a)¹⁸ by experiment.

The electrostatic analogy predicts that the interparticle force F between S- and D-particles and between D- and P-particles^{2,6} are dipolar-quadrupolar ones and F is proportional to $(r/a)^{-5}$. F between S- and P-particles is quadrupolar-quadrupolar one and exhibits the dependence of $(r/a)^{-6}$.

We have studied F between S- and D-, P- and D-, and S- and P- pairs as shown in Fig. 6. The magnitude of F is smaller than F for D-D pair. However, the general feature of the force curve is similar to D-D pair. F at large r can be fit well by a power of (r/a) . The best-fit curves are drawn in solid lines in Fig. 6. The obtained exponents respectively are -4.95, -5.04 and -5.78 for S-D, P-D and S-P pairs. They are in relatively good agreement with those predicted by electrostatic analogy. The value of (r/a) where F shows maximum value in magnitude is larger in D-S pair. This is due to the existence of the point defect which is distant from the surface of the particle and the repulsion between the particles is much larger than other pairs.

The dependence of F on the relative arrangement is different from the theoretical prediction of electrostatic analogy¹⁸. With approaching from the energetically unfavorable direction, the particles tend to rotate by themselves to reduce the interparticle energy. This makes the repulsive angular region smaller than that by the theoretical prediction at small r .

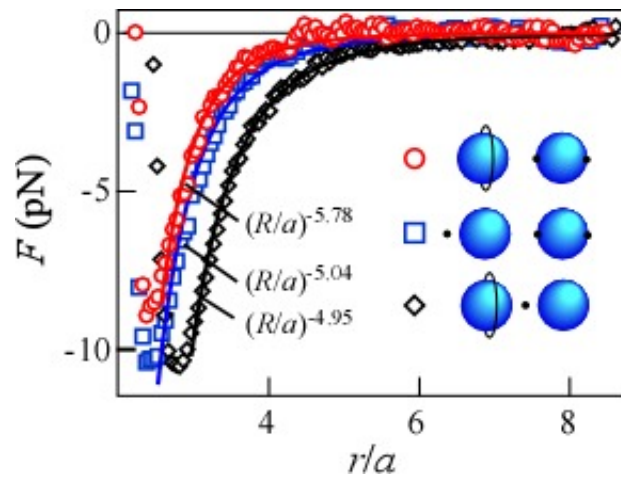


Figure 6. Dependence of the interparticle force F between different nematic colloidal pairs on the reduced interparticle distance r/a . The solid lines respectively are the best-fit ones of power of r/a .

4. CONSTRUCTION OF CLUSTERS OF NEMATIC COLLOIDS USING OPTICAL TWEEZERS

The nematic colloids limit its attractive direction and promote the appropriate directional connection between the particles. This largely differs from the isotropic connection between particles in simple liquid. Recently, the directional connection between colloidal particles has been achieved by utilizing the patched particles where the attractive interaction is limited to some special direction¹⁹⁻²¹. Such patched particles enable us to form open or empty spatial structures²¹. In nematic colloids, the particles are isotropic and the defect configuration governs this interaction.

The relative angular dependence of interparticle force in the D-D case is shown in Fig.5 (a). For parallel alignment of D-particles, the inline configuration is attractive and the side-by-side one is repulsive. Therefore, two particles tend to align inline and the chain cluster is stably formed. Such structure resembles a dipolar chain of colloids formed under electric and magnetic field. In this structure, the dipoles align in same direction as shown in Fig. 7 (a) and this can be called “ferro-chain”. When two dipoles align in antiparallel direction, “antiferro-chain” can be formed in the direction perpendicular to the alignment of liquid crystal as shown in Fig. 7 (b). By combining those two chains, one can build the colloidal clusters composed of straight lines²².

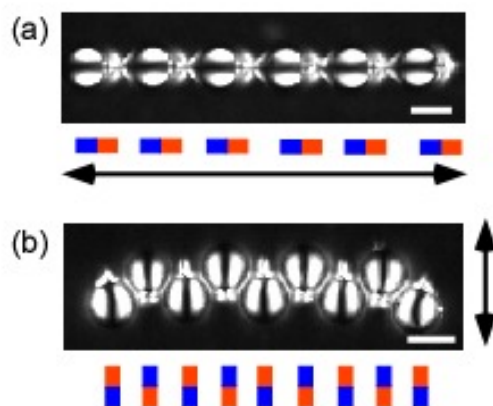


Figure 7. Polarizing microscope images of chains composed of D-particles under crossed polarizers. (a) ferro-chain. (b) antiferro-chain. The silica particles with $5\mu\text{m}$ in diameter are dispersed in pentyl-cyanobiphenyl (5CB). The scale bars are $5\mu\text{m}$. The arrows indicate the direction of alignment of 5CB. The schematic figures show the corresponding alignment of magnetic bars.

By utilizing the mixture of different sized D-particles, one can stably build more complex colloidal assemblies than those in simple liquid. In case of the isotropic attraction, the energetically stable structures are closely packed ones. However, the directional attractive and repulsive interaction can realize more extended or empty structures as energetically favored ones. We succeeded in building a dendritic assembly as shown in Fig. 8 with three different sized D-particles. The size of the particles we used in Fig. 8 are $10\mu\text{m}$, $5\mu\text{m}$ and $3\mu\text{m}$. Many smaller particles can joint with larger one within the attractive angular region of the larger one. The chains made of smaller particles repel each other and form an extended structure.

Moreover, in the case of the spherical colloids with the same size in two-dimension, the closest packing realizes hexagonal lattice. The directional interaction is able to control the type of two dimensional lattice. The various kinds of empty square lattice were reported in the binary mixture of D-and S-^{22,23} or S- and P-particles^{23,24}.

In the P-particles, the maximum attractive force appears in the oblique direction as shown in Fig. 5 (b). This particle has four bonds which can connect other P-particles. Therefore, one can construct the structure composed of empty rhombic lattice with P-particles with different sizes as shown in Fig. 9. The two inner angles of the basic unit is about 60° and 120° .

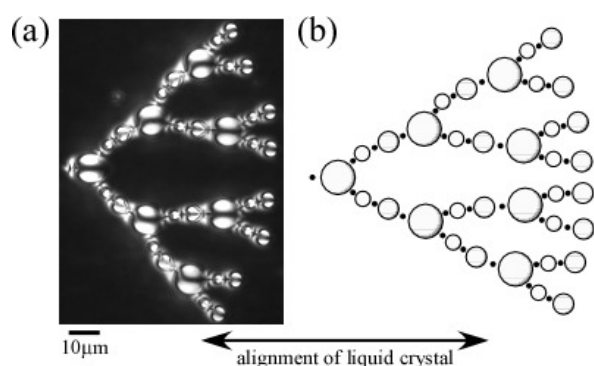


Figure 8. Cluster composed of dipolar nematic colloids with three different sizes. (a) Polarized microscope image under crossed polarizers, (b) Schematic alignment of colloids. The size of large, middle and small silica particles are 10 , 5 and $3\mu\text{m}$.

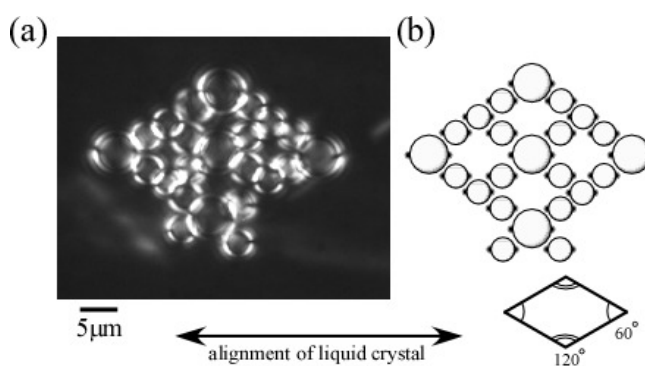


Figure 9. Cluster composed of planar quadrupolar nematic colloids with different sizes. (a) Polarized microscope image under crossed polarizers, (b) Schematic alignment of colloids. The inner angles of the basic unit are 60° and 120° . The size of the silica particles are 5 and $3\mu\text{m}$.

5. CONCLUSION

We measured the interparticle force between nematic colloids by some methods utilizing optical tweezers. By taking into account the hydrodynamic interaction between particles, the apparent difference in the obtained force by different methods has been reduced. The equivalency of the different methods extend the applicability of the measurement for more complex situation. The dependence of the interparticle force on the interparticle distance was measured for various nematic colloids with the same and different configuration. The information of the interaction helps to design the exotic assembled structures of colloids. The characteristic directional interaction has been applied to form unique colloidal clusters which cannot be realized in conventional spherical colloids in isotropic liquids. The concept of utilizing directional interparticle force to build colloidal crystal (cluster) can be extended to three-dimensions. This enables us to form various types of the colloidal crystals in three dimensions those are similar to the crystals of molecules and atoms.

ACKNOWLEDGEMENT

The authors thank Merck Japan Ltd. for providing the liquid crystal sample. They also thank Dr. J. Fukuda at AIST for valuable discussion on nematic colloids. This work was supported by Grant-in-Aid for Scientific Research, KAKENHI No. 23656052, 23340123 and 26287100 from JSPS and by Innovative Areas "Fluctuation & Structure" (Grant No. 25610125) from MEXT, Japan.

REFERENCES

- [1] Poulin, P., Stark, H., Lubensky, T. C. and Weitz, D. A., "Novel colloidal interactions in anisotropic fluids," *Science* 275, 1770-1773 (1997).
- [2] Lubensky, T. C., Pettey, D., Currier, N. and Stark, H., "Topological defects and interactions in nematic emulsions," *Phys. Rev. E* 57(1), 610-625 (1998).
- [3] Stark, H., "Physics of colloidal dispersions in nematic liquid crystals," *Phys. Rep.* 351(6), 387-474 (2001).
- [4] Fukuda, J., "Liquid crystal colloids: a novel composite material based on liquid crystals," *J. Phys. Soc. Jpn.* 78 (4), 041003 1-8 (2009).
- [5] Poulin, P. and Weitz, D. A., "Inverted and multiple nematic emulsions," *Phys. Rev. E* 57(1), 626-637 (1998).
- [6] Pergamenschik, V. M. and Uzunova, V. A., "Colloidal nematostatics," *Condens. Matter Phys.* 13(3), 33602 1-29 (2010).
- [7] Poulin, P., Cabuil, V. and Weitz, D. A., "Direct measurement of colloidal forces in an anisotropic solvent," *Phys. Rev. Lett.* 79(24), 4862-4865 (1997).
- [8] Skarabot, M., Ravnik, M., Zumer, S., Tkalec, U., Poberaj, I., Babic, D., Osterman, N. and Musevic, I., "Two-dimensional dipolar nematic colloidal crystals," *Phys. Rev. E* 76(5), 051406 1-8 (2007).
- [9] Noel, C. M., Bossis, G., Chaze, A.-M., Giulieri, F. and Lacis, S., "Measurement of elastic forces between iron colloidal particles in a nematic liquid crystal," *Phys. Rev. Lett.* 96(21), 217801 1-4 (2006).
- [10] Yada, M., Yamamoto, J. and Yokoyama, H., "Direct observation of anisotropic interparticle forces in nematic colloids with optical tweezers," *Phys. Rev. Lett.* 92(18), 185501 1-4 (2004).
- [11] Smalyukh, I. I., Lavrentovich, O. D., Kuzmin, A. N., Kachynski, A. V. and Prasad, P. N., "Elasticity-mediated self-organization and colloidal interactions of solid spheres with tangential anchoring in a nematic liquid crystal," *Phys. Rev. Lett.* 95(15), 157801 1-4 (2005).
- [12] Takahashi, K., Ichikawa, M. and Kimura, Y., "Direct measurement of force between colloidal particles in a nematic liquid crystal," *J. Phys.: Condens. Matter* 20(7), 075106 1-5 (2008).
- [13] Takahashi, K., Ichikawa, M. and Kimura, Y., "Force between colloidal particles in a nematic liquid crystal studied by optical tweezers," *Phys. Rev. E* 77(2), 020703(R) 1-4 (2008).
- [14] Loudet, J. C., Hanusse, P. and Poulin, P., "Stokes drag on a sphere in a nematic liquid crystal," *Science* 26, 1525 (2004).
- [15] Izaki, K. and Kimura, Y., "Hydrodynamic effects in the measurement of interparticle forces in nematic colloids," *Phys. Rev. E* 88(5), 054501 1-5 (2013).
- [16] Russel, W. B., Saville, D. A. and Schowalter, W. R., [Colloidal Dispersions], Cambridge University Press, Cambridge, (1989).
- [17] Kishita, T., Kondo, N., Takahashi, K., Ichikawa, M., Fukuda, J.-I. and Kimura, Y., "Interparticle force in nematic colloids: comparison between experiment and theory," *Phys. Rev. E* 84(2), 021704 1-9 (2011).

- [18] Izaki, K. and Kimura, Y., "Interparticle force between different types of nematic colloids," *Phys. Rev. E* 87(6), 062507 1-7 (2013).
- [19] Zhang, Z. and Glotzer, S. C., "Self-assembly of patchy particles," *Nano Letters* 4(8), 1407-1413 (2004).
- [20] Wang, Y., Breed, D. R., Manoharan, V. N., Feng, L., Hollingsworth, A. D., Weck, M. and Pine, D. J., "Colloids with valence and specific directional binding," *Nature* 491, 51-55 (2012).
- [21] Chen, Q., Bae, S. C. and Granick S., "Directed self-assembly of a colloidal kagome lattice," *Nature* 469, 381-384 (2011).
- [22] Kimura, Y., Kishita, T., Kita, K. and Kondo, N., "Nematic colloids– interaction between particles in anisotropic liquids," *J. Phys. Soc. Jpn.* 81 SA007 1-8 (2012).
- [23] Ognysta, U., Nych, A., Nazarenko, V., Musevic, I., Skarabot, M., Ravnik, M., Zumer, S., Poberaj, I. and Babic, D., "2D interactions and binary crystals of dipolar and quadrupolar nematic colloids," *Phys. Rev. Lett.* 100(21), 217803 1-4 (2008).
- [24] Ognysta, U., Nych, A., Nazarenko, V., Skarabot, M. and Musevic, I., "Design of 2D binary colloidal crystals in a nematic liquid crystal," *Langmuir* 25(20), 12092-12100 (2009).
- [25] Ognysta, U. M., Nych, A. B., Uzunova, V. A., Pergamenschik, V. M., Nazarenko, V. G., Skarabot M. and Musevic, I., "Square colloidal lattices and pair interaction in a binary system of quadrupolar nematic colloids," *Phys. Rev. E* 83(4), 041709 1-7 (2011).